

Modular Anticomputer Architecture and Cost Model

Compact Photon-Driven Antimatter Production System

1. Design Objective

Build a modular antimatter production system capable of exceeding CERN positron/antiproton output using photon-driven pair production rather than particle collision rings.

Target: - Container-scale unit - Megawatt-class electrical input - Microgram-scale positron production per year (scalable) - No circular accelerator infrastructure

2. System Architecture

Each Anticomputer Node (ACN) consists of:

1. High-Repetition Laser Wakefield Accelerator (LWFA)
2. High-Z Microstructured Pair-Production Target
3. Gamma Collimation and Cascade Chamber
4. Magnetic Positron Separation System
5. Penning or Ioffe Trap Storage Array
6. Cryogenic + Power Conditioning Module
7. Control + Compute Integration Layer

Physical footprint: ~12m x 2.5m x 2.5m (shipping container scale)

3. Performance Assumptions (2035 Realistic)

Laser System: - 5 MW continuous electrical input - 20% wall-plug efficiency (advanced solid-state systems) - Effective gamma production efficiency: 5% - Pair production conversion: 1% - Capture efficiency: 30%

Net antimatter energy conversion: $0.20 \times 0.05 \times 0.01 \times 0.30 = 0.000003 = 0.0003\%$

Total annual electrical input: $5 \text{ MW} \times 31,536,000 \text{ s} \approx 1.58 \times 10^{14} \text{ J}$

Antimatter energy equivalent per year: $1.58 \times 10^{14} \text{ J} \times 0.000003 \approx 4.7 \times 10^8 \text{ J}$

Mass equivalent: $E = mc^2 \Rightarrow m = E / c^2 \approx 4.7 \times 10^8 / (9 \times 10^{16}) \approx 5.2 \times 10^{-9} \text{ kg} \approx 5 \text{ micrograms per year}$

Result: $\sim 5 \mu\text{g}$ positrons per year per node

4. Capital Cost Estimate (Per Node)

High-power laser system: \$40M LWFA vacuum + beamline: \$10M Target chamber + shielding: \$8M Magnetic separation system: \$12M Cryogenics + traps: \$15M Power conditioning + cooling: \$10M Control systems + integration: \$5M Facility buildout + safety: \$20M

Estimated Total CapEx per Node: $\sim \$120\text{M}$

5. Operational Costs (Annual)

Electricity (5 MW @ \$0.05/kWh): $5 \text{ MW} \times 8760 \text{ h} = 43,800 \text{ MWh}$ Cost $\approx \$2.19\text{M}$ per year

Staff + maintenance: \$5M Consumables + target replacement: \$2M

Total OpEx $\approx \$9\text{--}10\text{M}$ per year

6. Cost per Microgram

Annual output: $5 \mu\text{g}$ Annual OpEx: $\sim \$10\text{M}$

Operating cost per $\mu\text{g} \approx \$2\text{M}$

Amortized CapEx over 20 years: $\$120\text{M} / 20 = \6M per year

Total annual cost $\approx \$16\text{M}$

Effective cost per $\mu\text{g} \approx \$3.2\text{M}$

7. Scaling Strategy

Deploy 20 nodes:

Capital cost $\approx \$2.4\text{B}$ Total output $\approx 100 \mu\text{g}$ per year

Deploy 200 nodes:

Capital cost $\approx \$24\text{B}$ Total output $\approx 1 \text{ mg}$ per year

This exceeds CERN positron-scale output while remaining modular and distributed.

8. Competitive Advantage Over CERN

- No 27 km ring
 - No TeV proton collisions
 - Lower civil infrastructure cost
 - Modular expansion
 - Direct positron generation
 - Compatible with integrated computational hardware
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9. Economic Viability

Primary early markets: - Deep-space propulsion ignition research - Medical isotope spin-offs - High-energy physics research - National strategic energy reserve

Secondary long-term use: - Antimatter-catalyzed fusion systems

10. Conclusion

A fully operational anticomputer node capable of microgram-scale antimatter production can be built for approximately \$120M capital cost and ~\$10M annual operating cost. Scaled networks can exceed collider-based production at significantly reduced infrastructure complexity.

This makes antimatter production an industrial engineering problem rather than a mega-collider problem.